

- Solve all the problems below in **2 hours and 30 minutes**.
- **Mobile phones are not allowed in the examination hall.**
- Please justify all your answers through proof or counterexample.
- Scoring: **1,2** $\rightarrow$ 4 points, **3,4** $\rightarrow$ 6 points.

1. Let a, b be real numbers. If $a \geq b > 0$, show that

$$\frac{1}{8a}(b-a)^2 \leq \frac{a+b}{2} - \sqrt{ab} \leq \frac{1}{8b}(b-a)^2.$$

2. Solve the following equations in $\mathbb{R}$

$$E(5x) = 4E(x), \quad \text{and} \quad E(2x+3) = E(3x+1).$$

3. We consider the numerical sequence $(u_n)_{n \in \mathbb{N}}$ defined by :

$$\begin{cases} u_0 = a, & a \in \mathbb{R}^+, \\ u_{n+1} = \frac{1}{2}u_n(u_n + 1), & n \in \mathbb{N}. \end{cases}$$

Part I: We assume that: $a = \frac{1}{2}$.

- Show that, $\forall n \in \mathbb{N} : 0 < u_n < 1$.
- Study the monotonicity of $(u_n)_{n \in \mathbb{N}}$.
- Deduce that $(u_n)_{n \in \mathbb{N}}$ is convergent. What is its limit?

Part II: Assume that: $a > 1$.

- Show that, $\forall n \in \mathbb{N} : u_n > 1$.
- Show that $(u_n)_{n \in \mathbb{N}}$ is increasing, and that $u_n \geq \left(\frac{1+a}{2}\right)^n$ for all $n \in \mathbb{N}$.
- Deduce the nature of $(u_n)_{n \in \mathbb{N}}$.

4. Consider the real function f given by :

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0, \\ 0 & x = 0. \end{cases}$$

- (a) Study the continuity of f on its domain of definition.
- (b) Study the differentiability of f at 0.
- (c) State the Mean Value Theorem.

Consider the function $g(x) = \arctan(2x + 1)$.

- (1) Determine the domain of definition of g .
- (2) Calculate the derivative of g and specify its domain of existence.

Correction

1. (2 points to each side) We have

$$\frac{a+b}{2} - \sqrt{ab} = \frac{(\sqrt{a} - \sqrt{b})^2}{2} = \frac{(b-a)^2}{2(\sqrt{a} + \sqrt{b})^2}$$

as $a \geq b > 0$, then $2\sqrt{a} \geq \sqrt{a} + \sqrt{b} \geq 2\sqrt{b}$ so $\frac{1}{8a} \leq \frac{1}{2(\sqrt{a} + \sqrt{b})^2} \leq \frac{1}{8b}$

which implies that $\frac{(b-a)^2}{8a} \leq \frac{(b-a)^2}{2(\sqrt{a} + \sqrt{b})^2} \leq \frac{(b-a)^2}{8b}$ then we get

$$\forall a, b : a \geq b > 0, \frac{1}{8a} (b-a)^2 \leq \frac{a+b}{2} - \sqrt{ab} \leq \frac{1}{8b} (b-a)^2.$$

2. (2 points to each equation) Done in tutoring (voir votre cahier de TD)
3. We consider the numerical sequence $(u_n)_{n \in \mathbb{N}}$ defined by:

$$\begin{cases} u_0 = a, & a \in \mathbb{R}^+ \\ u_{n+1} = \frac{1}{2}u_n(u_n + 1), & n \in \mathbb{N} \end{cases}$$

Part I: We assume that: $a = \frac{1}{2}$.

- (a) (1 point) Show that, $\forall n \in \mathbb{N} : 0 < u_n < 1$

By recurrence:

- For $n = 0$, we have: $u_0 = \frac{1}{2}$ and $0 < \frac{1}{2} \leq 1$ is true.
 - Suppose: $0 < u_n < 1$ and show $0 < u_{n+1} \leq 1$, for all $n \in \mathbb{N}$.
- we have:

$$0 < u_n < 1, \text{ then } \begin{cases} 0 < \frac{1}{2}u_n \leq \frac{1}{2} \\ 1 < u_{n+1} < 2 \end{cases} \quad \text{so: } 0 < \frac{1}{2}u_n(u_n + 1) < 1$$

Hence: $\forall n \in \mathbb{N} : 0 < u_n < 1$

(b) (1 point) Study the monotonicity of $(u_n)_{n \in \mathbb{N}}$

We have

$$\begin{aligned} u_{n+1} - u_n &= \frac{1}{2}u_n(u_n + 1) - u_n \\ &= \frac{1}{2}u_n(u_n - 1) \dots \dots \dots \end{aligned}$$

From question 1: $0 < u_n < 1$, then: $\begin{cases} u_n > 0 \\ u_n - 1 < 0 \end{cases}$ then: $u_{n+1} - u_n < 0$

Hence the sequence $(u_n)_{n \in \mathbb{N}}$ is increasing

(c) (1/2 point+1/2 point) Deduce that $(u_n)_{n \in \mathbb{N}}$ is convergent. What is its limit?

The sequence $(u_n)_n$ is decreasing and lower bound, then $(u_n)_n$ converges to its lower bound $\lim_{n \rightarrow \infty} (u_n)_n = l$.

On the other hand, we have $u_{n+1} = f(u_n)$, $f : x \mapsto \frac{1}{2}x(x+1)$ continuous, then l verifies the relation:

$$\begin{aligned} f(l) = l &\Rightarrow \frac{1}{2}l(l+1) = l \Rightarrow \frac{1}{2}l(l-1) = 0 \\ &\Rightarrow l = 0 \text{ ou } l = 1, \end{aligned}$$

like sequence is decreasing $(u_n)_n$ $u_n < \dots < u_1 < u_0 = \frac{1}{2}$ and $0 < u_n < 1$, then $l = 0$

Part II: Assume that: $a > 1$.

(a) (1 point) Show that, $\forall n \in \mathbb{N} : u_n > 1$

By recurrence

- For $n = 0$, we $a : u_0 = a > 1$ is true.
- Assume: $u_n > 1$ and show $u_{n+1} > 1$, for all $n \in \mathbb{N}$. We have:

$$u_n > 1, \text{ then } \begin{cases} \frac{1}{2}u_n > \frac{1}{2} \\ u_n + 1 > 2 \end{cases} \text{ so: } \frac{1}{2}u_n(u_n + 1) > 1$$

Hence: $\forall n \in \mathbb{N} : u_n > 1$.

(b) (1 point=1/2+1/2) Show that $(u_n)_{n \in \mathbb{N}}$ is increasing and that $u_n \geq \left(\frac{1+a}{2}\right)^n$ for all $n \in \mathbb{N}$

- All terms of the sequence $(u_n)_n$ are strictly positive, so:

$$\frac{u_{n+1}}{u_n} = \frac{1}{2}(u_n + 1) > 1.$$

Hence $(u_n)_n$ is increasing.

- Let us show that $u_n \geq \left(\frac{1+a}{2}\right)^n$ for all $n \in \mathbb{N}$, by induction:

For $n = 0$, we have $u_0 = a > 1$. Now suppose $u_n \geq \left(\frac{1+a}{2}\right)^n$, let's show that $u_{n+1} \geq \left(\frac{1+a}{2}\right)^{n+1}$.

For $u_0 = a > \left(\frac{1+a}{2}\right)^0 = 1$, and since the sequence is increasing, then for all $n \in \mathbb{N}$, we have $u_n \geq u_0 = a$ and according to the hypothesis $u_n \geq \left(\frac{1+a}{2}\right)^n$. This result gives:

$$\begin{aligned} u_{n+1} &= \frac{1}{2}u_n(u_n + 1) \\ &\geq \frac{1}{2}\left(\frac{1+a}{2}\right)^n(a+1) \\ &\geq \left(\frac{1+a}{2}\right)^{n+1} \end{aligned}$$

Hence: $\forall n \in \mathbb{N} : u_n \geq \left(\frac{1+a}{2}\right)^n$.

- (c) (1 point) Deduce the nature of $(u_n)_{n \in \mathbb{N}}$

As

$$\lim_{n \rightarrow +\infty} u_{n+1} \geq \lim_{n \rightarrow +\infty} \underbrace{\left(\frac{1+a}{2}\right)^{n+1}}_{>1} = +\infty$$

So $\lim_{n \rightarrow +\infty} u_n = +\infty$, hence: $(u_n)_n$ is divergent.

4. Consider the real function f given by:

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

- (a) (1 point) Study the continuity of f on its domain of definition.

$$\begin{aligned} D_f &= \mathbb{R}^* \cup \{0\} \\ &= \mathbb{R} \end{aligned}$$

– On $\mathbb{R}^*$, we have

$x \mapsto x \sin\left(\frac{1}{x}\right)$ continuous on $\mathbb{R}^*$ (product and composition of continuous functions)

– At point $x_0 = 0$, we have

$$f \text{ is continuous at } 0 \Leftrightarrow \lim_{x \rightarrow 0} f(x) = f(0).$$

$$f(0) = 0$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0 \text{ because } : \left| \sin\left(\frac{1}{x}\right) \right| \leq 1 \text{ and } \lim_{x \rightarrow 0} x = 0$$

Hence f is continuous at 0, which implies that f is continuous on $\mathbb{R}$.

(b) (1 point) Study the differentiability of f at 0

f is differentiable at 0 $\Leftrightarrow \lim_{x \rightarrow 0} \frac{f(x)-f(0)}{x} = l, \quad l \in \mathbb{R}.$

Then:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} &= \lim_{x \rightarrow 0} \frac{x \sin\left(\frac{1}{x}\right)}{x} \\ &= \lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right) \\ &= \text{does not exist.} \end{aligned}$$

Hence f is not differentiable at point $x_0 = 0$.

(c) (1 point) State the Mean Value Theorem. Let $f : [a, b] \rightarrow \mathbb{R}$ be a function, if:

- f is continuous on $[a, b]$
- f is differentiable on $]a, b[$

then

$$\exists c \in]a, b[, \text{ such as: } f(b) - f(a) = (b - a)f'(c).$$

Consider the function $g(x) = \arctan(2x + 1)$

(1) (1 point) Determine the domain of definition of g .

$$D_g = \mathbb{R}.$$

(2) (1 point)+(1 point) Calculate the derivative of g by specifying its domain of existence g is differentiable on $\mathbb{R}$, (because composed of functions $x \mapsto 2x + 1$ differentiable on $\mathbb{R}$ and $x \mapsto \arctan x$ differentiable on $\mathbb{R}$).

$$\begin{aligned} g'(x) &= (\arctan(2x + 1))' \\ &= \frac{2x}{1 + (2x + 1)^2} \\ &= \frac{2}{4x^2 + 2x + 2}, \quad x \in \mathbb{R}. \end{aligned}$$